DDA3020 Machine Learning
Lecture 02 Probability and Information Theory

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Outline

1. Probability, event, random variables
2. Probability of discrete random variables
3. Probability of continuous random variables
4. Some popular distributions
5. Information theory

1. Probability, event, random variables
2. Probability of discrete random variables
3. Probability of continuous random variables
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5. Information theory

Random experiment, sample space, event

- Random experiment: we describe a random experiment by its **procedure** and observations of its **outcomes**. For example, we toss a coin 2 times, and observe which side is up after each toss.
- Sample space: All possible outcomes of the random experiment form a sample space S. For the above coin toss example, we define

$$S = \{(Head, Head), (Head, Tail), (Tail, Head), (Tail, Tail)\}.$$

T_1 T_2 $|S| = 4$

- Event: A **subset** of sample space S, denoted as A, can be called as an event in a random experiment, *i.e.*, $A \subset S$. In the above example, we define an event A as *at least one head up*, then it can be represented by

$$A = \{(Head, Head), (Head, Tail), (Tail, Head)\} \subset S.$$

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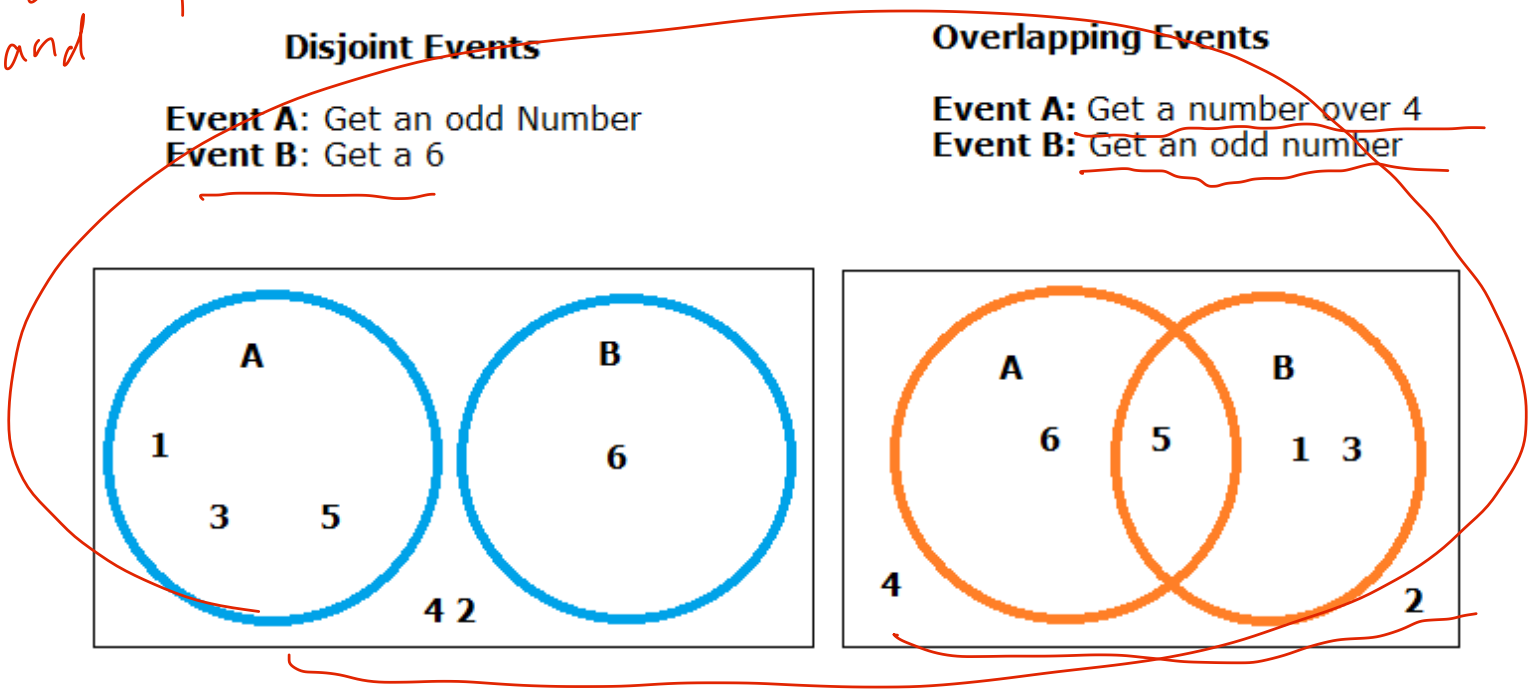
Probability of events

Assuming events $A \subset S$ and $B \subset S$, the probabilities of events related with and must satisfy,

- $P(A) \geq 0$
- $P(S) = 1$
- If $A \cap B = \emptyset$, then $P(A \cup B) = P(A) + P(B)$; otherwise, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

cup → or

cap → and



Random variables

- A random variable is a real valued function from the sample space S to a real space $\mathbb{R}$, as follows:

$$X : S \to \mathbb{R}$$

- Still take the 2-times coin toss as an example, if we define the random variable as the number of tails, then we have

$$X((H,H)) = 0, X((H,T)) = 1, X((T,H)) = 1, X((T,T)) = 2.$$

  Then, the output space of X is denoted as $\{0,1,2\}$, also called state space $\mathcal{X}$.

- There are two types of random variables:
  - **Discrete**: $\mathcal{X}$ is discrete
  - **Continuous**: $\mathcal{X}$ is continuous

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Probability of discrete random variables

- Probability of discrete random variable describes the chance of each state x in $\mathcal{X}$ for random variable X in a random experiment, denoted as

$$P(X = x), x \in \mathcal{X}.$$

- It should satisfy

$$P(X = x) \geq 0, \forall x \in \mathcal{X}; \sum_{x \in \mathcal{X}} P(X = x) = 1.$$

Probability of discrete random variables

- Joint probability of two random variables X and Y:

$$P(X = x, Y = y) = P(X = x|Y = y)P(Y = y) = P(Y = y|X = x)P(X = x)$$

- Marginal probability of X and Y:

$$P(X = x) = \sum_{y \in \mathcal{Y}} P(X = x, Y = y)$$

$$P(Y = y) = \sum_{x \in \mathcal{X}} P(X = x, Y = y)$$

- Conditional probability:

$$P(X = x|Y = y) = P(X = x, Y = y)/P(Y = y)$$

$$P(Y = y|X = x) = P(X = x, Y = y)/P(X = x)$$

Bayes rule and application on medical diagnosis

- Bayes Rule: Combining the definition of conditional probability with the product and sum rules yields Bayes rule, as follows:

$$P(X = x|Y = y) = \frac{P(X = x, Y = y)}{P(Y = y)} = \frac{P(X = x)P(Y = y|X = x)}{\sum_{x' \in \mathcal{X}} P(X = x')P(Y = y|X = x')} \tag{1}$$

- Suppose that you do a medical test for breast cancer, the test result could be *positive* or *negative*. We denote $x = 1$ as the event of a positive test and $x = 0$ as the event of a negative test. We denote $y = 1$ as the event of having breast cancer and $y = 0$ as the event of no breast cancer.
- Suppose that if one has breast cancer, the test will be positive with the probability 0.8, *i.e.*,

$$P(x = 1|y = 1) = 0.8. \tag{2}$$

- Then, if one gets a positive test result, what is the probability of having breast cancer? $P(y = 1|x = 1) = 0.8$?

Bayes rule and application on medical diagnosis

- It is WRONG! It ignores the prior probability of having breast cancer.
- According to statistics, the average risk of a woman in the United States developing breast cancer sometime in her life is about 13%, *i.e.*,

$$P(y = 1) = 0.13. \tag{3}$$

- We also need to take into account the fact that the test may be a **false positive** or **false alarm**. Unfortunately, such false positives are quite likely (with current screening technology):

false alarm rate

$$P(x = 1|y = 0) = 0.1. \tag{4}$$

- Combining all the above probabilities using the Bayes rule, we can compute

$$P(y = 1|x = 1) = \frac{P(x = 1|y = 1)P(y = 1)}{P(x = 1|y = 1)P(y = 1) + P(x = 1|y = 0)P(y = 0)}$$

$$= \frac{0.8 \times 0.13}{0.8 \times 0.13 + 0.1 \times 0.87} = 0.5445. \tag{5}$$

$P(Y|X) = \frac{P(Y,X)}{P(X)}$

It tells that if you test positive, you have about a 54% chance of really having breast cancer!

Independent random variables

X: your math score Z: you like apple

Y: your ML score $X \perp Y$

- Independent: If X and Y are independent, denoted as $X \perp Y$, then the joint probability can be represented as the product of two marginals, *i.e.*,

$$X \perp Y \iff P(X,Y) = P(X)P(Y).$$

- Given the above independence, we can use fewer parameters to define a joint probability. Suppose that X has 3 states, Y has 4 states, then we need $3-1=2$ and $4-1=3$ free parameters to define $P(X)$ and $P(Y)$, respectively.
- If without independence, how many free parameters do we need to define the joint probability $P(X,Y)$? $(3\times 4)-1=11$.
- If given the independence, *i.e.*, $P(X,Y)=P(X)P(Y)$, how many free parameters do we need? $(3-1)+(4-1)=5$.

$P(X,Y) = P(X)P(Y)$

$P(X=x_1, Y=y_3) = P(X=x_1)P(Y=y_3)$

Expectation and variance of discrete random variables

- Expectation (or mean): $E[X] = \sum_{x \in \mathcal{X}} xP(X = x)$
- Expectation of a function: $E[f(X)] = \sum_{x \in \mathcal{X}} f(x)P(X = x)$
- Moments: expectation of power of X: $M_k = E[X^k]$
- Variance: Average (squared) fluctuation from the mean

$$\mathrm{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2 = M_2 - M_1^2. \tag{6}$$

- Standard deviation: Square root of variance, *i.e.*,

$$\mathrm{Std} = \sqrt{\mathrm{Var}(X)}. \tag{7}$$

- `Exercise` : For the random variable of the number of tails in the random experiment of 2-times coin toss, please compute the above values.

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Continuous random variables

- A random variable X is continuous if its state space $\mathcal{X}$ is uncountable.
- In this case, $P(X = x) = 0$ for each x.
- If $p_X(x)$ is a probability density function (PDF) for X, then

$$P(a < X < b) = \int_a^b p(x)dx \tag{8}$$

$$P(a < X < a + dx) \approx p(a) \cdot dx \tag{9}$$

- The cumulative distribution function (CDF) is $F_X(x) = P(X \leq x)$. We have that $p_X(x) = F'(x)$, and $F(x) = \int_{-\infty}^{x} p(s)ds$.

Bivariate continuous distributions: Marginalization, Conditioning, and Independence

- $p_{X,Y}(x,y)$, joint probability density function of X and Y
- $\int_x \int_y p(x,y)dxdy = 1$
- Marginal distribution: $p(x) = \int_{-\infty}^{\infty} p(x,y)dy$
- Conditional distribution: $p(x|y) = \frac{p(x,y)}{p(y)}$
- Note: $P(Y = y) = 0$! Formally, conditional probability in the continuous case can be derived using infinitesimal events.
- Independence: X and Y are independent if $p_{X,Y}(x,y) = p_X(x)p_Y(y)$

Expectation and variance of continuous random variables

Similar to that of discrete random variables, only change the summation $\sum$ to the integral $\int$.

- Expectation (or mean): $\mu = E[X] = \int_{\mathcal{X}} x \cdot p(x)dx$
- Moments: expectation of power of X: $M_k = E[X^k] = \int_{\mathcal{X}} x^k \cdot p(x)dx$
- Variance: Average (squared) fluctuation from the mean

$$\mathrm{Var}(X) = E[(X - E[X])^2] = E[X^2] - E[X]^2 = M_2 - M_1^2. \tag{10}$$

- Standard deviation: Square root of variance, *i.e.*,

$$\mathrm{Std} = \sqrt{\mathrm{Var}(X)}. \tag{11}$$

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Binary variables (discrete r.v.)

- We first consider the probability of a binary random variable $x \in \{0, 1\}$. Suppose that you toss a coin, and $x = 1$ denotes the event of 'heads', while $x = 0$ indicates the event of 'tails'.
- The probability of $x = 1$ is described by a parameter μ,

$$p(x = 1|\mu) = \mu, \tag{12}$$

where $\mu \in [0, 1]$, and we can obtain that $p(x = 0|\mu) = 1 - \mu$.

- The probability distribution over x can therefore be written in the form

$$\text{Bern}(x|\mu) = \mu^x (1 - \mu)^{1-x}, \tag{13}$$

which is called Bernoulli distribution.

- Its mean and variance are

$$E[x] = \sum_x x \text{Bern}(x|\mu) = \mu, \tag{14}$$

$$\text{var}[x] = E[(x - \mu)^2] = \mu(1 - \mu) \tag{15}$$

Binomial variables

- Imagine that you toss the coin N times, and each tossing follows the Bernoulli distribution $p(x|\mu)$. We denote the variable m as the number of heads, then its distribution is formulated as follows:

$$\text{Bin}(m|N,\mu) = \binom{N}{m}\mu^m(1-\mu)^{N-m}, \tag{16}$$

which is called Binomial distribution, where

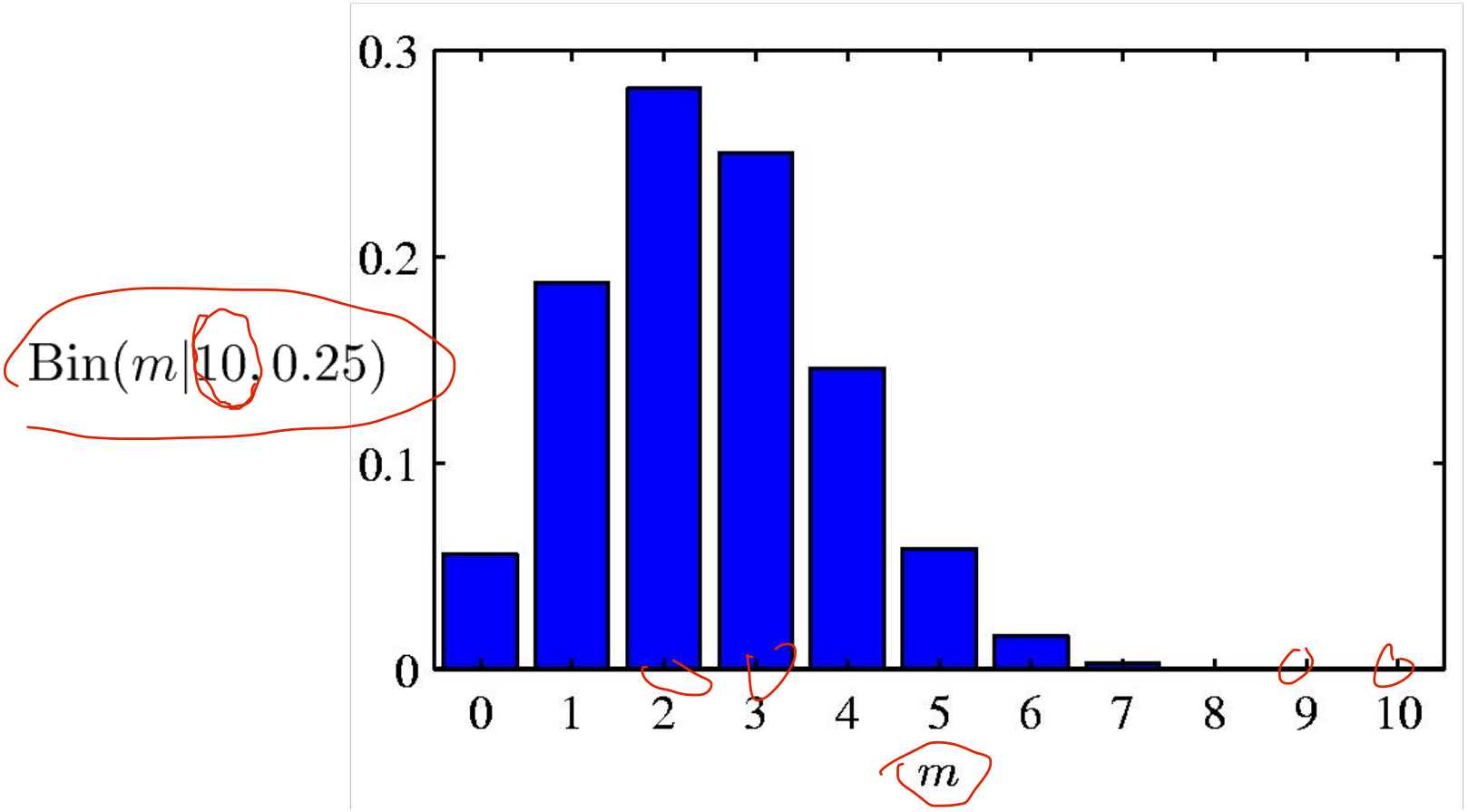
$$\binom{N}{m} = \frac{N!}{(N-m)!m!}. \tag{17}$$

- Its mean and variance are

$$E[m] = \sum_{m=0}^{N} m\text{Bin}(m|N,\mu) = N\mu, \tag{18}$$

$$\text{var}[m] = E[(m-N\mu)^2] = N\mu(1-\mu). \tag{19}$$

Binomial distribution



Gaussian distribution (continuous r.v.)

- The Gaussian, also known as the **normal** distribution, is a widely used model for the distribution of **continuous** variables. In the case of a single variable x, the Gaussian distribution can be written in the form

$$\mathcal{N}(x|\mu,\sigma) = \frac{1}{(2\pi\sigma^2)^{\frac{1}{2}}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right), \tag{20}$$

  where μ is the **mean** and σ^2 is the **variance**.

- For a D-dimensional vector $\boldsymbol{x}$, the multivariate Gaussian distribution takes the form

$$\mathcal{N}(\boldsymbol{x}|\boldsymbol{\mu},\boldsymbol{\Sigma}) = \frac{1}{(2\pi)^{\frac{D}{2}}|\boldsymbol{\Sigma}|^{\frac{1}{2}}} \exp\left(-\frac{(\boldsymbol{x}-\boldsymbol{\mu})^\top \boldsymbol{\Sigma}^{-1}(\boldsymbol{x}-\boldsymbol{\mu})}{2}\right), \tag{21}$$

  where $\boldsymbol{\mu}$ is a D-dimensional **mean vector**, and $\boldsymbol{\Sigma}$ is a $D \times D$ **covariance matrix**, and $|\boldsymbol{\Sigma}|$ denotes the determinant of $\boldsymbol{\Sigma}$.

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What is information

- To know what is information, we need to know a bit who is Claude Shannon and his information theory (let's see a video).
- Shannon defined the measure that quantifies the uncertainty of an event with given probability - a bit.
- For a discrete random variable (a source) with a finite alphabet, as follows

$$\mathcal{X} = \{x_0, \cdots, x_k, \cdots x_S\},$$

where the probability of each symbol is given by $P(X = x_k) = p_k$.

- We define the information as

$$I(x_k) = \log \frac{1}{p_k} = -\log(p_k).$$

If the logarithm is base 2, information is given in bits.

- Note that $I(x_k) \geq 0$, *i.e.*, non-negative. The equality only holds when $p_k = 1$, which means there is **no uncertainty**, then **no information**.

What is information

- It represents the *surprise* of seeing the outcome (a highly probable outcome is not surprising).

event	probability	surprise
one equals one	1	0 bits
wrong guess on a 4-choice question	3/4	0.415 bits
correct guess on true-false question	1/2	1 bit
correct guess on a 4-choice question	1/4	2 bits
seven on a pair of dice	6/36	2.58 bits
win any prize at Euromilhões	1/24	4.585 bits
win Euromilhões Jackpot	$\approx$ 1/76 million	$\approx$ 26 bits
gamma ray burst mass extinction today	$< 2.7 \cdot 10^{-12}$	$>$ 38 bits

Larger surprise/uncertainty, more information/bits.

Entropy

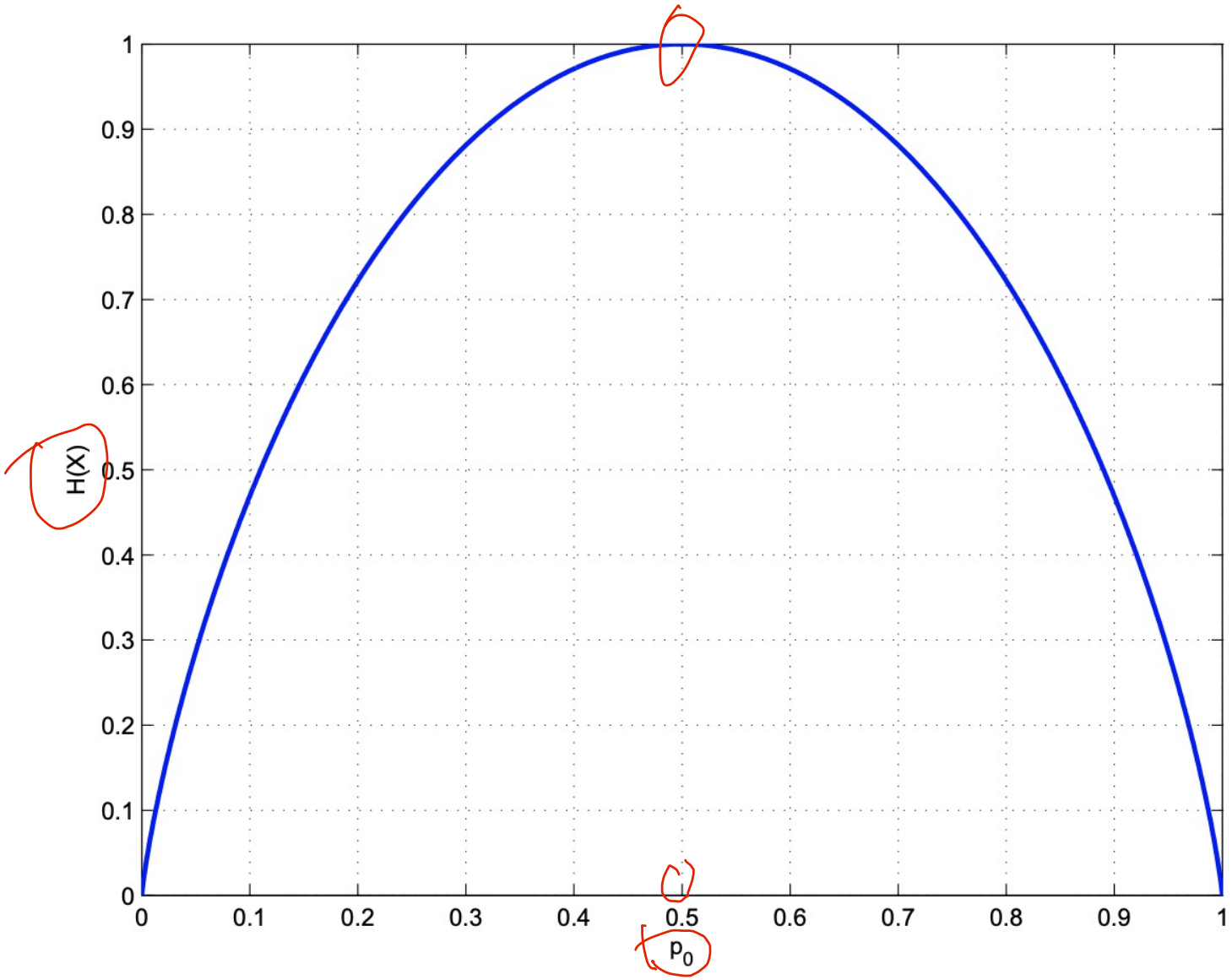
- Entropy is defined as the **expected value of information** from a source,

$$\begin{aligned} H_P(\mathcal{X}) &= E[I(x_k)] \\ &= \sum_{x_k \in \mathcal{X}} p_k \cdot I(x_k) \\ &= -\sum_{x_k \in \mathcal{X}} p_k \cdot \log(p_k). \end{aligned}$$

- Let $\mathcal{X} = \{0, 1\}$ be a binary source with p_0 and p_1 being the probability of symbols $x_0 = 0$ and $x_1 = 1$, respectively, we have

$$\begin{aligned} H_P(\mathcal{X}) &= E[I(x_k)] \\ &= -p_0 \log p_0 - p_1 \log p_1 \\ &= -p_0 \log p_0 - (1 - p_0) \log (1 - p_0) \end{aligned}$$

Entropy of binary source



Cross Entropy

- Cross-entropy builds upon the idea of entropy. It calculates the number of bits required to represent or transmit an average event from one distribution $P(X)$, compared to another distribution $Q(X)$,

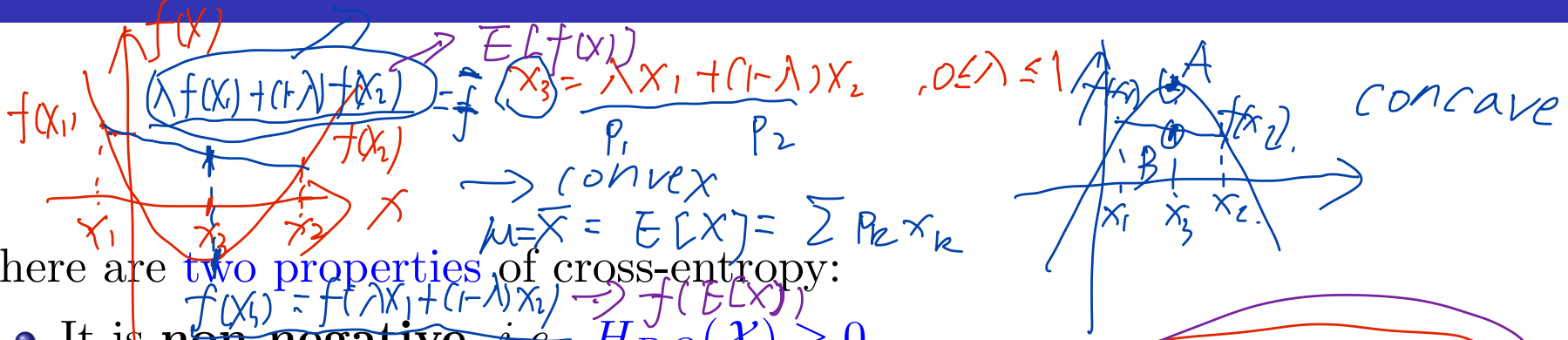
$$H_{P,Q}(\mathcal{X}) = -\sum_{x_k \in \mathcal{X}} P(X = x_k) \cdot \log(Q(X = x_k)),$$

  where $P(X = x_k)$ is the probability of the event x_k in $P(X)$, $Q(X = x_k)$ is the probability of event x_k in $Q(X)$.

- Let $\mathcal{X} = \{0, 1\}$ be a binary source with p_0 and p_1 being the probability of symbols $x_0 = 0$ and $x_1 = 1$, respectively, then we have

$$\begin{aligned} H_{P,Q}(\mathcal{X}) &= -p_0 \log q_0 - p_1 \log q_1 \\ &= -p_0 \log q_0 - (1 - p_0) \log (1 - q_0). \end{aligned}$$

Cross Entropy



There are two properties of cross-entropy:

- It is **non-negative**, *i.e.*, $H_{P,Q}(\mathcal{X}) \geq 0$
- Cross-entropy is not smaller than entropy, *i.e.*, $H_{P,Q}(\mathcal{X}) \geq H_P(\mathcal{X})$, and the equality holds only when $P = Q$.

Exercise: Prove the above two properties
(hint: Jensen's inequality and $\log(\cdot)$ is concave)

Reference: https://machinelearningmastery.com/cross-entropy-for-machine-learning/

Relative entropy: Kullback-Leibler divergence

- The relative entropy between two discrete probability distributions P and Q is

$$D_{P,Q}(\mathcal{X}) = \sum_{x_k \in \mathcal{X}} P(X = x_k) \cdot \log \left(\frac{P(X = x_k)}{Q(X = x_k)} \right),$$

- The relative entropy between two continuous probability density functions $p_X(x)$ and $q_X(x)$ is defined as follows

$$D_{P,Q}(\mathcal{X}) = \int_{x \in \mathcal{X}} p_X(x) \log \left(\frac{p_X(x)}{q_X(x)} \right) dx$$

- It is also called KL divergence, which measures the distance between two distributions.

$$KL(P \| Q) = \int c_1 \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right) \log \frac{c_1 \exp\left(-\frac{(x-\mu_1)^2}{2\sigma_1^2}\right)}{c_2 \exp\left(-\frac{(x-\mu_2)^2}{2\sigma_2^2}\right)} dx$$

Relative entropy: Kullback-Leibler divergence

- Exercise 1: What is a KL divergence for a discrete r.v. ?
- There are two properties of KL divergence:
  - **Non-negativity**: $D_{P,Q}(\mathcal{X}) \geq 0$
  - **Asymmetry**: $D_{P,Q}(\mathcal{X}) \neq D_{Q,P}(\mathcal{X})$
- Exercise 2: Prove the above two properties (hint: Jensen inequal and $\log(\cdot)$ is concave)
- The cross-entropy $H_{P,Q}(\mathcal{X})$ is the entropy of the distribution $H_P(\mathcal{X})$ plus the additional KL divergence $D_{P,Q}(\mathcal{X})$.

$$H_{P,Q}(\mathcal{X}) = H_P(\mathcal{X}) + D_{P,Q}(\mathcal{X})$$

- Exercise 3: Can you prove the equality of the equation?

Reference:

https://machinelearningmastery.com/cross-entropy-for-machine-learning/

https://stats.stackexchange.com/questions/335197/why-kl-divergence-is-non-negative